

Md Rafi Ahmed

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About Me

Driven Computer Science student specializing in full-stack architecture, intelligent automation tools, and AI-driven platforms. Demonstrated capability in building scalable applications, ranging from autonomous web agents to comprehensive Learning Management Systems integrating generative AI and serverless databases. Eager to leverage strong problem-solving skills in modern web technologies, deterministic logic, and UI/UX design to engineer innovative, high-performance software solutions.

Skills

- **Languages & Core Concepts:** C, C++, JavaScript, Python, HTML/CSS, DSA, Object-Oriented Programming
- **Backend & Databases:** Node.js, Express.js, PostgreSQL (Neon Serverless), RESTful APIs
- **Tools & Technologies:** Google Gemini API, Git, GitHub, Playwright, JWT Authentication, RBAC, Cloudinary

Projects

New Dawn Companion | Full-Stack LMS

Node.js, Express, PostgreSQL, Gemini API

- Architected a comprehensive Learning Management System featuring a dynamic CMS landing page, live teacher gradebooks, and role-based access control (RBAC) using JWT authentication.
- Integrated Google Gemini 3.5 Flash API to autonomously parse raw text files into structured, incrementally numbered sets of 25-question automated quizzes.
- Designed a robust relational database using Neon Serverless PostgreSQL to manage class hierarchies, study material hosting, and secure user data.
- Built a highly secure exam environment with anti-cheat mechanisms, including tab-switch auto-submission protocols, disabled keyboard shortcuts, and real-time countdown timers.

Appointo | Doctor Appointment Portal

Full-Stack Web Development

- Developed a full-stack healthcare platform with secure patient registration, authentication, and live appointment booking functionality.
- Implemented features for schedule management, doctor availability tracking, and comprehensive patient history with a highly responsive, accessible user interface.
- Engineered robust database integration for CRUD operations, ensuring a seamless and fast user experience.

WebPilot | Autonomous Voice-Enabled Web Agent

FastAPI, Playwright, AI Integration

- Architecting an autonomous, voice-activated web assistant framework using FastAPI, Playwright, and HTML/CSS/JS.
- Developing backend routing to facilitate real-time web automation and task execution across browser environments.
- Integrating advanced AI APIs to process natural language commands and translate them into deterministic browser actions.

Smart Automated Garage System

Arduino Uno, C++, Hardware Automation

- Built a fully autonomous, bidirectional smart gate system applying custom logic for seamless vehicle detection and entry control.

- Integrated and programmed an Arduino Uno with dual HC-SR04 ultrasonic sensors and an SG90 servo motor for precise mechanical actuation.

C Inventory Management System

C, File I/O, Data Architecture

- Developed a lightweight, console-based inventory management application utilizing core C programming concepts.
- Implemented strict CRUD functionality utilizing standard binary file operations (‘.dat’) for fast, local data persistence.
- Engineered robust memory management and NULL pointer checks to prevent segmentation faults during edge-case scenarios.

Technical Practice & Implementation

Programming Fundamentals (C, DSA)

- Have a strong grip on basic console programs using conditional statements, loops, functions, and arrays in C.
- Practiced problem-solving using recursion, sorting algorithms, and user-defined functions.
- Implemented core data structures like arrays and basic linked lists.

Beginner Python Practice

- Built simple console applications and scripts to grasp Python syntax, loops, and logic.
- Performed file input/output operations for reading, parsing, and writing data efficiently.

Education

B.Tech in Computer Science and Engineering (2025-2029)

Pursuing

JIS College of Engineering, Kalyani

Interest

- Participating in hackathons and building scalable full-stack applications
- High-contrast, cinematic Web Design & UI/UX Architecture
- Exploring core computer science concepts and serverless database integration